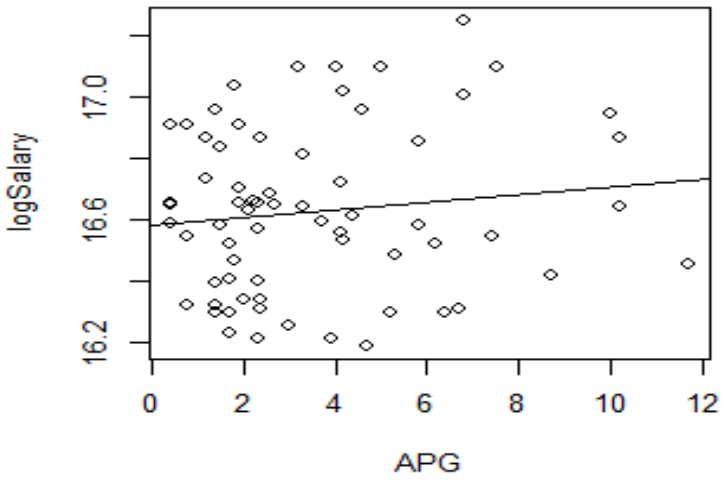
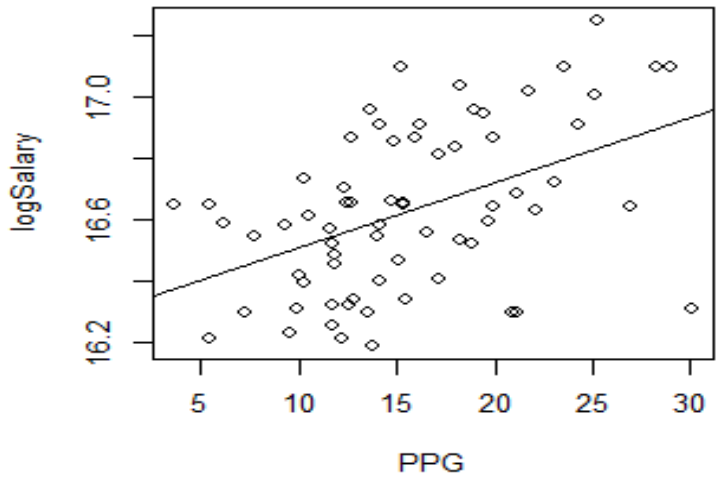
Multiple Linear regression

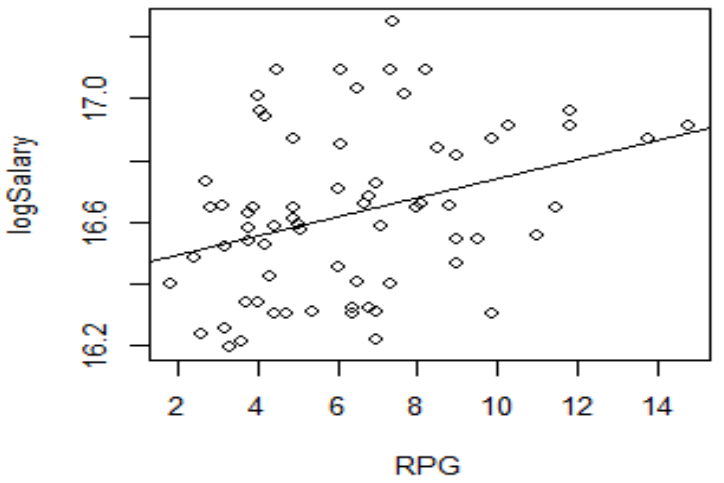
Example: What determine an NBA player's salary?

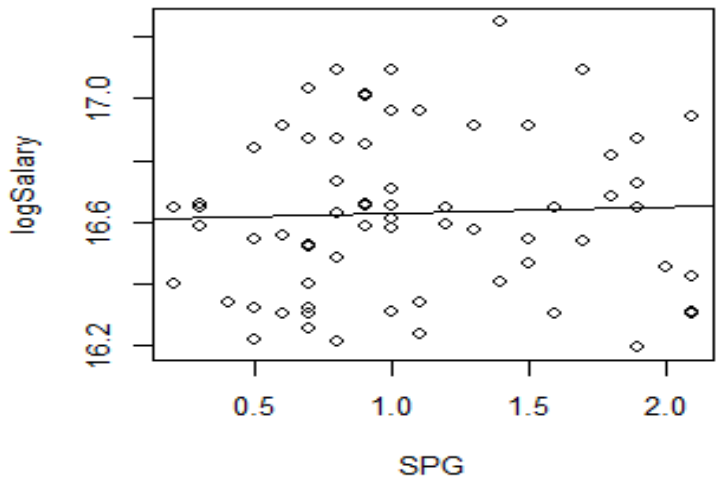
The data contain salary of 68 NBA players for their 2016 contracts and some key statistics associated with each of these players including assists per game (APG), points per game (PPG), rebounds per game (RPG), and steals per game (SPG). Denote the four predictors by X1, X2, X3, X4. Let Y be the log(Salary).




logSalary
16.2
16.6
17.0
0
2
4
6
8
10
12
APG


logSalary
16.2
16.6
17.0
5
10
15
20
25
30
PPG


logSalary
16.2
16.6
17.0
2
4
6
8
10
12
14
RPG


logSalary
16.2
16.6
17.0
0.5
1.0
1.5
2.0
SPG

Model Formulation

- The multiple linear regression model is:

$$Y_i = \beta_0 + \beta_1 X1_i + \beta_2 X2_i + \beta_3 X3_i + \beta_4 X4_i + \varepsilon_i$$

  - β_i's are called partial regression coefficients

- Model assumptions:
  - $E(\varepsilon_i) = 0$
  - ε_i's have constant variance, i.e., $Var(\varepsilon_i) = \sigma^2$
  - ε_i's are independent and normally distributed

- The above assumptions imply that

$$E(Y_i) = \beta_0 + \beta_1 X1_i + \beta_2 X2_i + \beta_3 X3_i + \beta_4 X4_i$$

$$Var(Y_i) = \sigma^2$$

Interpretation of the regression coefficients

β_0	*is the expected logSalary for a player whose APG, PPG, RPG and SPG are all equal to 0*
β_1	*is the change in the expected logSalary by having one more APG while holding PPG, RGP and SPG constant*
β_2	*is the change in the expected logSalary by having one more PPG while holding APG, RGP and SPG constant*
β_3	*is the change in the expected logSalary by having one more RPG while holding APG, PGP and SPG constant*
β_4	*is the change in the expected logSalary by having one more SPG while holding APG, PGP and RPG constant*

The Method of Least Squares

- Define the ***deviation***: $Y_i - (\beta_0 + \beta_1 X1_i + \beta_2 X2_i + \beta_3 X3_i + \beta_4 X4_i)$
- Define the ***sum of squared deviations***:

$$Q = \sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 X1_i - \beta_2 X2_i - \beta_3 X3_i - \beta_4 X4_i)^2$$

- Find $\widehat{\beta_i} = b_i, i = 0, \cdots, 4$ such that Q is minimized.
- The fitted sample regression line is

$$\hat{Y} = b_0 + b_1 X1 + \cdots + b_4 X4$$

Example: What determine a NBA player's salary?

- Write out the sample regression line and interpret the regression coefficients

```
> m<-lm(logSalary~APG+PPG+RPG+SPG)
> m$coefficients
(Intercept)         APG         PPG         RPG         SPG
16.17933159  0.02353490  0.02401581  0.03039068 -0.18805587
```

Matrix Representation

Write $Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, X = \begin{pmatrix} 1 & X1_1 & \cdots & X4_1 \\ 1 & X1_2 & \cdots & X4_2 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X1_n & \cdots & X4_n \end{pmatrix},$

$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_4 \end{bmatrix}, \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}, b = \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_4 \end{bmatrix}$

Then $Y = X\beta + \epsilon$ and $b = \left(X^T X\right)^{-1} X^T Y$, X is often referred as the design matrix.

I_n is an $n \times n$ **identity** matrix; its diagonal elements are all 1 and off-diagonal elements are all 0.

Consider two $n \times n$ matrices A and B, if $AB = I_n$, then we say B is the inverse of A; we write $B = A^{-1}$.

Things to look at

- How can the model help explain the variability in the data (ANOVA)?
- What can we say about the regression coefficients (confidence intervals, hypothesis testing)?
- How do we obtain confidence and prediction intervals for the response?
- Does the fitted model fit the data well? Are any model assumptions being violated (residual diagnostics)?
- If there is a problem, what do we do (transformations etc)?
- How do we determine a "good" model?

ANOVA – partitioning of squares

$$SSTO = \sum_{i=1}^{n} (Y_i - \bar{Y})^2$$ *Sum of squared total*

$$SSE = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$$ *Sum of squared error*

$$SSR = \sum_{i=1}^{n} (\hat{Y}_i - \bar{Y})^2$$ *Sum of squared regression*

The Coefficient of Multiple Determination

$$R^2 = \frac{SSR}{SSTO} = 1 - \frac{SSE}{SSTO} \qquad 0 \le R^2 \le 1$$

measures the proportional reduction in the variability in Y with the use of X1, X2, ……,X(p-1) as predictor variables, where (p-1) is the number of predictors.

Caution: adding more variables to the model will only increase R^2!!

Adjusted R^2: $R^2_{adj} = 1 - \frac{SSE/(n-p)}{SSTO/(n-1)}$, where *n* is the sample size.

ANOVA Table

Source	DF	SS	MS	F	P
Regression	p-1	SSR	SSR/p-1	MSR/MSE	
Error	n-p	SSE	SSE/n-p		
Total	n-1	SSTO	SSTO/n-1		

We use this F statistic to test the overall regression relationship.

$$H_0: \beta_1 = \beta_2 = \cdots = \beta_{p-1} = 0$$

$$H_a: at\ least\ one\ \beta_k \neq 0$$

Rejects for large F statistics as quantified by the p-value. If the p-value is less than the level of significance, then reject, otherwise do not reject.

Note: If you reject in multiple linear regression, you do not know which or how many are not zero. You only know that at least one is not zero.

Example: What determine an NBA player's salary?

- The function summary() gives the R^2, adjusted R^2, and the F-test result.

```
> summary(m)

Call:
lm(formula = logSalary ~ APG + PPG + RPG + SPG)

Residuals:
    Min      1Q  Median      3Q     Max
-0.5193 -0.1637  0.0278  0.1587  0.4031

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 16.179332   0.097586 165.796  < 2e-16 ***
APG          0.023535   0.014652   1.606  0.11323
PPG          0.024016   0.005507   4.361 4.89e-05 ***
RPG          0.030391   0.010104   3.008  0.00378 **
SPG         -0.188056   0.073967  -2.542  0.01348 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2185 on 63 degrees of freedom
Multiple R-squared:  0.368,     Adjusted R-squared:  0.3279
F-statistic: 9.171 on 4 and 63 DF,  p-value: 6.646e-06
```

Inference about β_k: confidence interval

- Confidence interval: $b_k \pm t_{\alpha/2,n-p}\, s(b_k)$
  - b_k: estimated regression coefficient
  - $s(b_k)$: standard error of b_k
  - n: sample size
  - p: number of regression coefficients
  - $t_{\alpha/2,n-p}$: the t-value that cuts off α/2 area to its right for a t distribution with (n-p) degrees of freedom

Inference about β_k: hypothesis testing

- Hypotheses testing:

$$H_0: \beta_k = 0 \quad vs \quad H_a: \beta_k \neq 0$$

- The test statistic $t = \frac{b_k}{s(b_k)}$ follows a t distribution with (*n-p*) degrees of freedom
- This test is for the marginal value of including Xk in the model provided X1, X2,……,X(k-1),X(k+1),……,X(p-1) are already included.

- If you **reject**, you are concluding that Xk does provide additional explanatory value given the other variables are already included.

- If you **fail to reject** you are concluding that Xk does not provide additional explanatory value given the other variables are included in the model.

- If the p-value is smaller than the level of significance, reject H_0 , otherwise fail to reject H_0 .

Example: What determine an NBA player's salary?

- Obtain and interpret the 95% confidence intervals for the regression coefficients.

```
> confint(m,level=.95)
                    2.5 %       97.5 %
(Intercept) 15.984321624 16.37434156
APG         -0.005745358  0.05281516
PPG          0.013011199  0.03502042
RPG          0.010198886  0.05058247
SPG         -0.335867081 -0.04024466
```

- Test whether the regression coefficients are significantly different from zero. Interpret your results.

```
> summary(m)

Call:
lm(formula = logSalary ~ APG + PPG + RPG + SPG)

Residuals:
    Min      1Q  Median      3Q     Max
-0.5193 -0.1637  0.0278  0.1587  0.4031

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 16.179332   0.097586 165.796  < 2e-16 ***
APG          0.023535   0.014652   1.606  0.11323
PPG          0.024016   0.005507   4.361 4.89e-05 ***
RPG          0.030391   0.010104   3.008  0.00378 **
SPG         -0.188056   0.073967  -2.542  0.01348 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Confidence and prediction intervals

- Consider a new data point taking values $X1_h, X2_h, \cdots, X(p-1)_h$, the fitted value for this new data point is given by

$$\hat{Y}_h = b_0 + b_1 X1_h + \cdots + b_{p-1} X(p-1)_h$$

- Confidence interval for the mean response:

$$\hat{Y}_h \pm t_{\alpha/2,n-p}\, s(\hat{Y}_h)$$

  - $s(\hat{Y}_h)$: standard error estimate of $\hat{Y}_h$

- Prediction interval for a new observation:

$$\hat{Y}_h \pm t_{\alpha/2,n-p}\, s(Y_{h(new)} - \hat{Y}_h)$$

  - $s(Y_{h(new)} - \hat{Y}_h)$: standard error estimate of $Y_{h(new)} - \hat{Y}_h$

Example: What determine an NBA player's salary?

- Dwyane Wade's 2015 statistics were

```
> X<-data.frame(Player,APG,PPG,RPG,SPG)
> X[which(Player=="Dwyane Wade"),]
        Player APG PPG RPG SPG
9 Dwyane Wade 4.6  19 4.1 1.1
```

- Estimate the average Salary for a player with the same performance statistics.

```
> newdata<-data.frame(APG=4.6,PPG=19,RPG=4.1,SPG=1.1)
> conf<-predict(m,newdata,interval="confidence",level=.95)
> conf
       fit      lwr      upr
1 16.66163 16.58234 16.74092
> exp(conf) #convert to dollar scale
       fit      lwr      upr
1 17220874 15908166 18641904
```

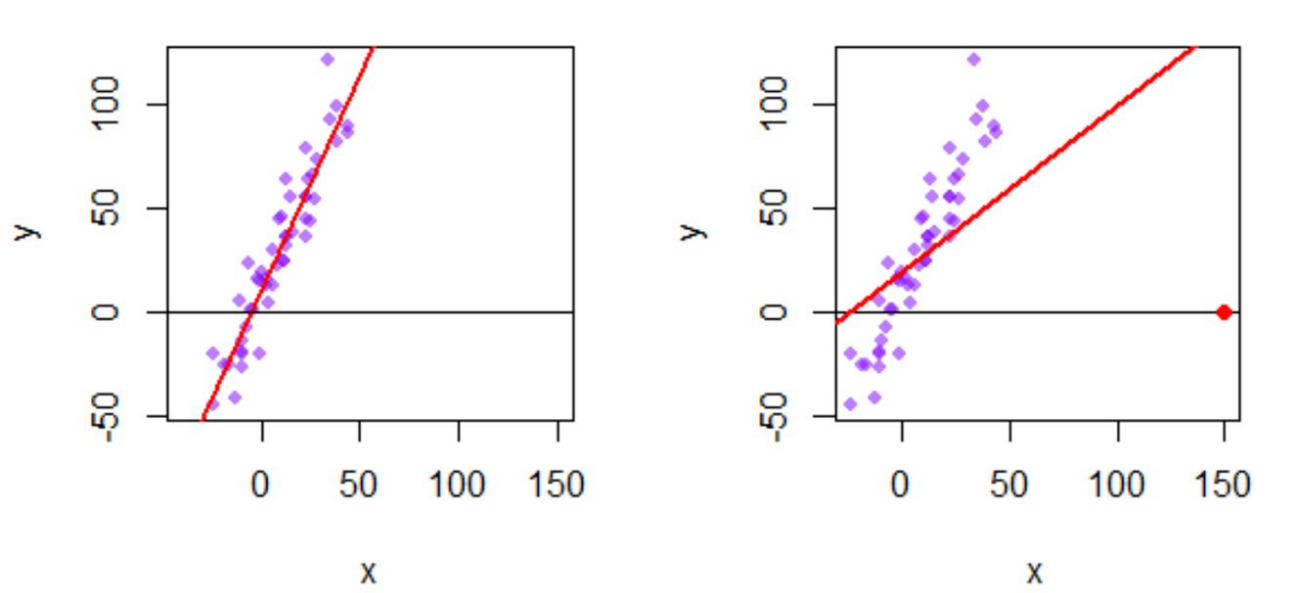
- Obtain a prediction interval for the actual Salary for a player with the same statistics.

```
> pred<-predict(m,newdata,interval="prediction",level=.95)
> pred
       fit      lwr      upr
1 16.66163 16.21795 17.10532
> exp(pred) #convert to dollar scale
       fit      lwr      upr
1 17220874 11050091 26837653
```

- Dwyane Wade's actual Salary for 2016 is 23,200,000. Do you think that he got a good contract? The Heat initially offered him 10,000,000 for the 2016. Would that have been a fair offer based on his past performance?

Diagnostics about the predictors

- Each data point tries to pull the sample regression line towards itself.
- Points with extreme predictor values (e.g. the red point in 2nd graph) have more leverage.

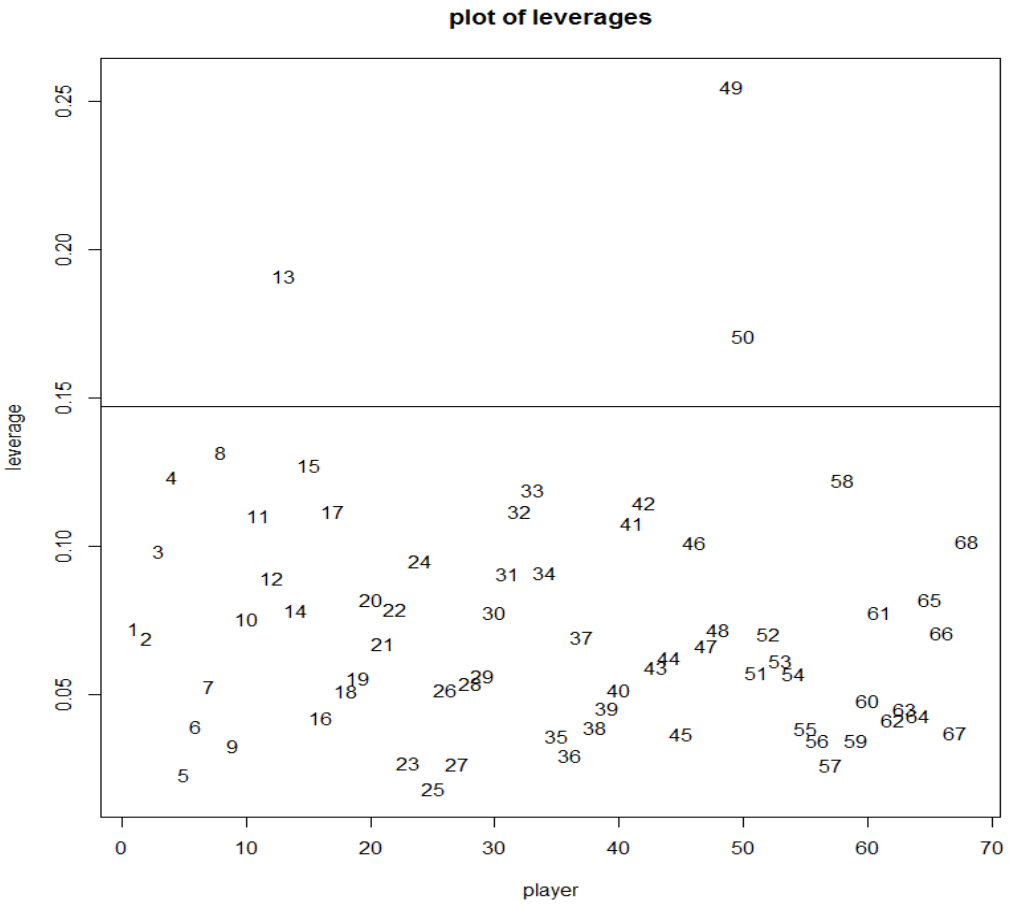


- How to find such points with many predictors?

Diagnostics about the predictors

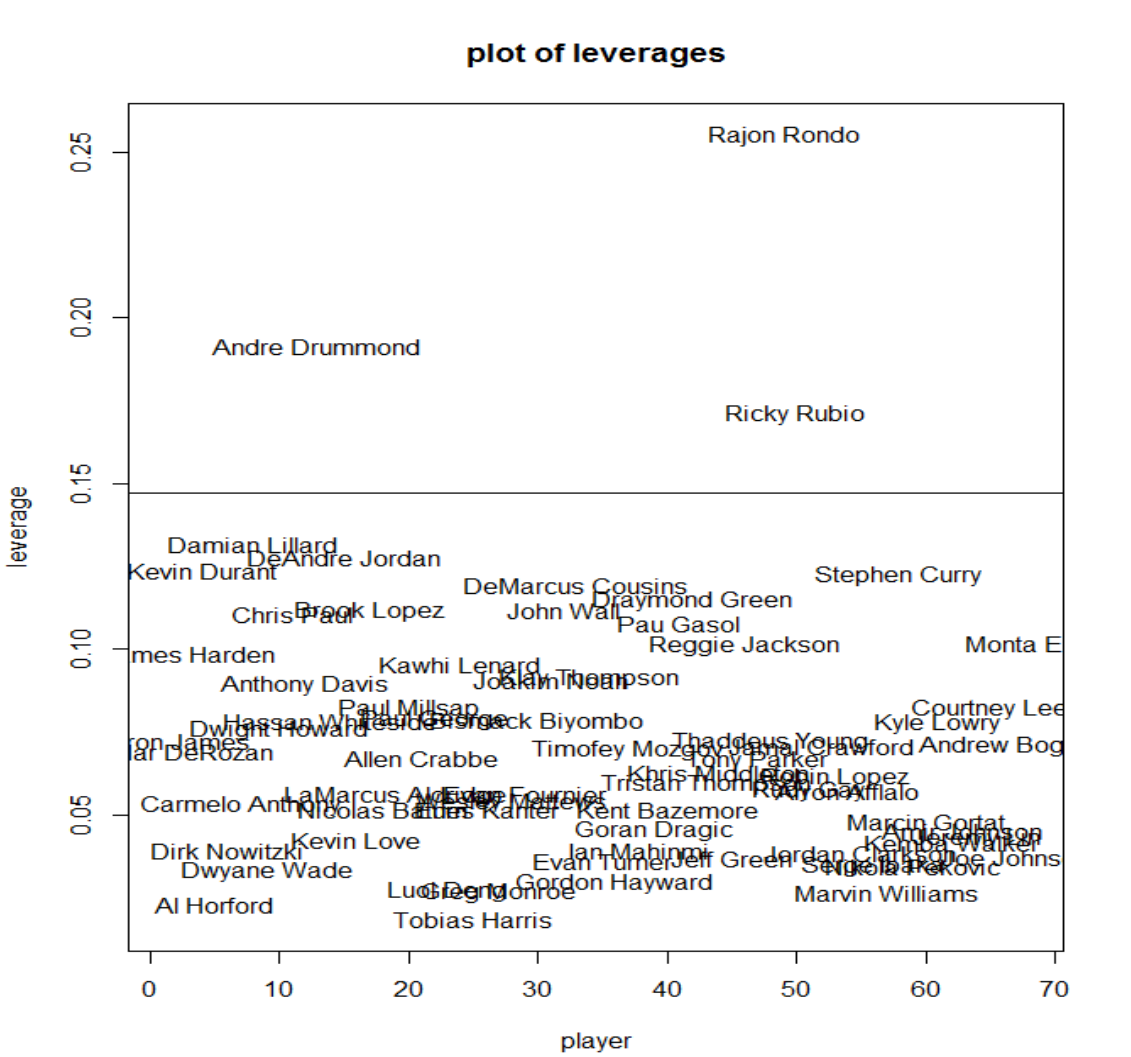
- A **high leverage point** is an outlier in terms of the predictor values.
- The ith diagonal element of the so-called Hat matrix, H_i, is a measure of the distance between the X value for the ith case and the average of the X values for all n cases.
- H_i is called the leverage for the ith case.
- So, a large value of H_i can indicate a high leverage point, which means this case is distant from the center of all X observations.
- It is known that $0<H_i<1$ and $\sum_{i=1}^{n} H_i = p$, so a point is considered to have a high leverage when H_i is larger than $(2p)/n$ (=0.147 for the NBA Salary example).

```
> leverage<-hatvalues(m)
> threshhold<-2*5/length(Player)
> player<-1:length(Player)
> plot(player,leverage,main="plot of leverages",type="n")
> text(player,leverage,labels=player)
> abline(h=threshhold)
```



	Player	Salary	APG	PPG	RPG	SPG
13	Andre Drummond	22116750	0.8	16.2	14.8	1.5
49	Rajon Rondo	14000000	11.7	11.9	6.0	2.0
50	Ricky Rubio	13550000	8.7	10.1	4.3	2.1

```
> plot(player,leverage,main="plot of leverages",type="n")
> text(player,leverage,labels=Player)
> abline(h=threshhold)
```



Detect Extrapolation Beyond the Range of the Data (not required)

- It can be shown that $var(\hat{Y}_i) = \sigma^2 H_i$. Thus, there is a larger variability in $\hat{Y}_i$ if the i[th] data point's predictor values are extreme.
- Let X be the design matrix
- Let $X_h = (1, X1_h, \cdots, X(p-1)_h)^T$ be a column vector of predictor values at a new level h. It can be shown

$$var(\hat{Y}_h) = \sigma^2 (X_h)^T (X^T X)^{-1} X_h$$

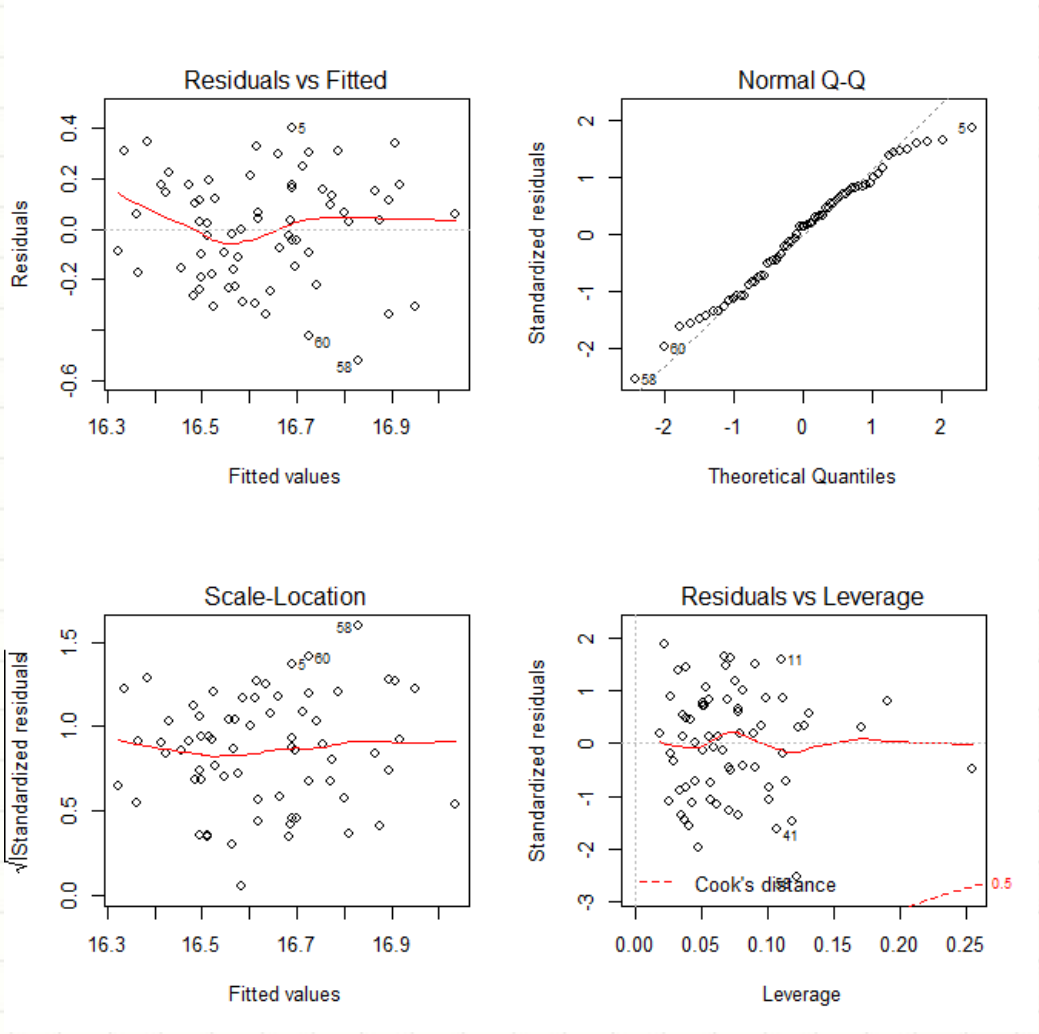
- A large value of $(X_h)^T (X^T X)^{-1} X_h$ indicates that the estimate $\hat{Y}_h$ may not be reliable, which can be used as evidence for possible extrapolation beyond the range of the data.

Residual Diagnostics

- If the multiple linear regression model is appropriate (use Residuals vs Fitted)
- If the errors have constant variance (use Residuals vs Fitted and/or Scale-Location)
- If the errors are independent (plot residuals against time)
- If there are outliers present (use Residuals vs Fitted and/or Residuals vs Leverage)
- If the errors are normally distributed (use Normal Q-Q plot)
- If there are any influential points (use Cook's distance)

Example: What determine an NBA player's salary?

```
> par(mfrow=c(2,2))
> plot(m)
```

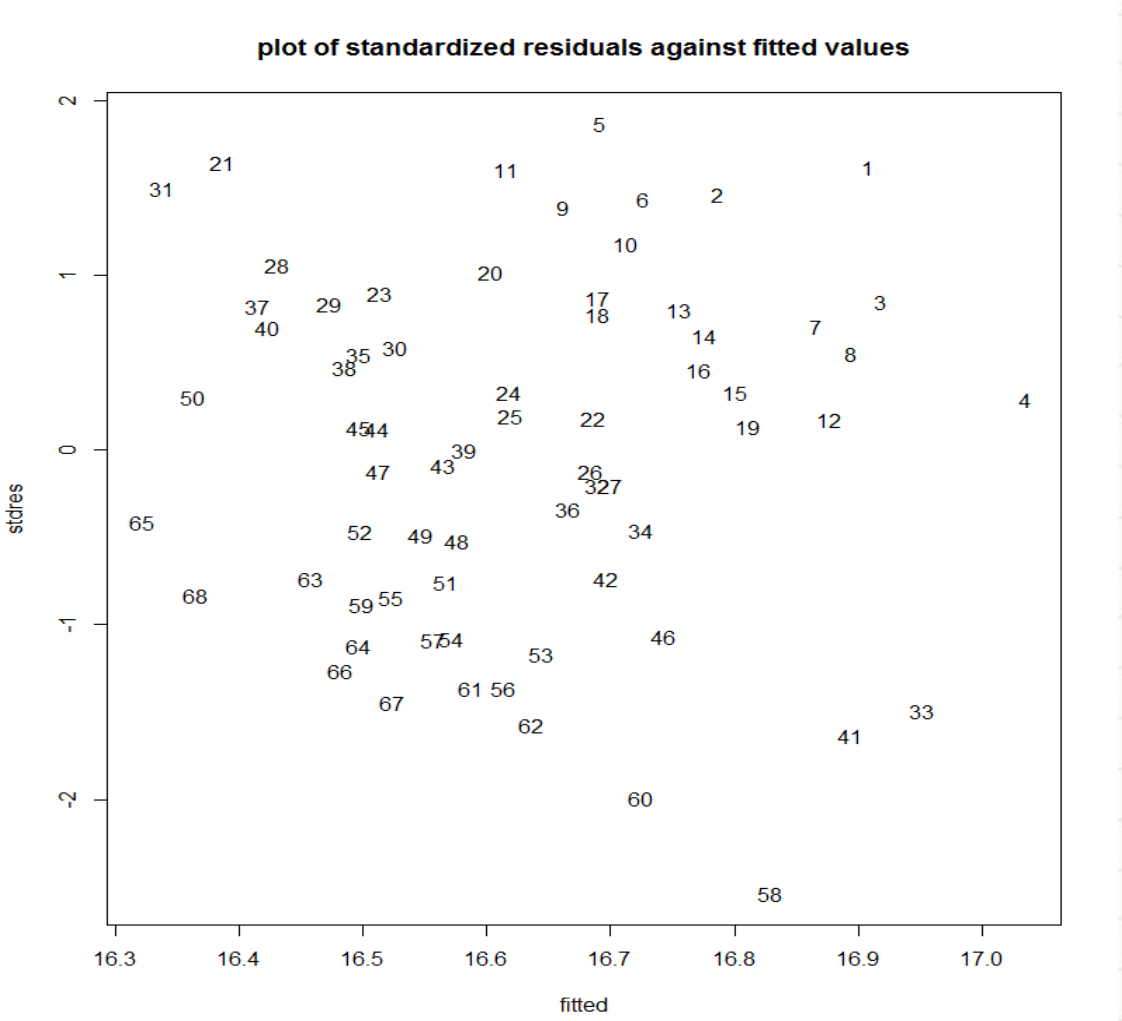


- Comment on whether the various model assumptions are satisfied based on the residual plots shown in the previous slide.

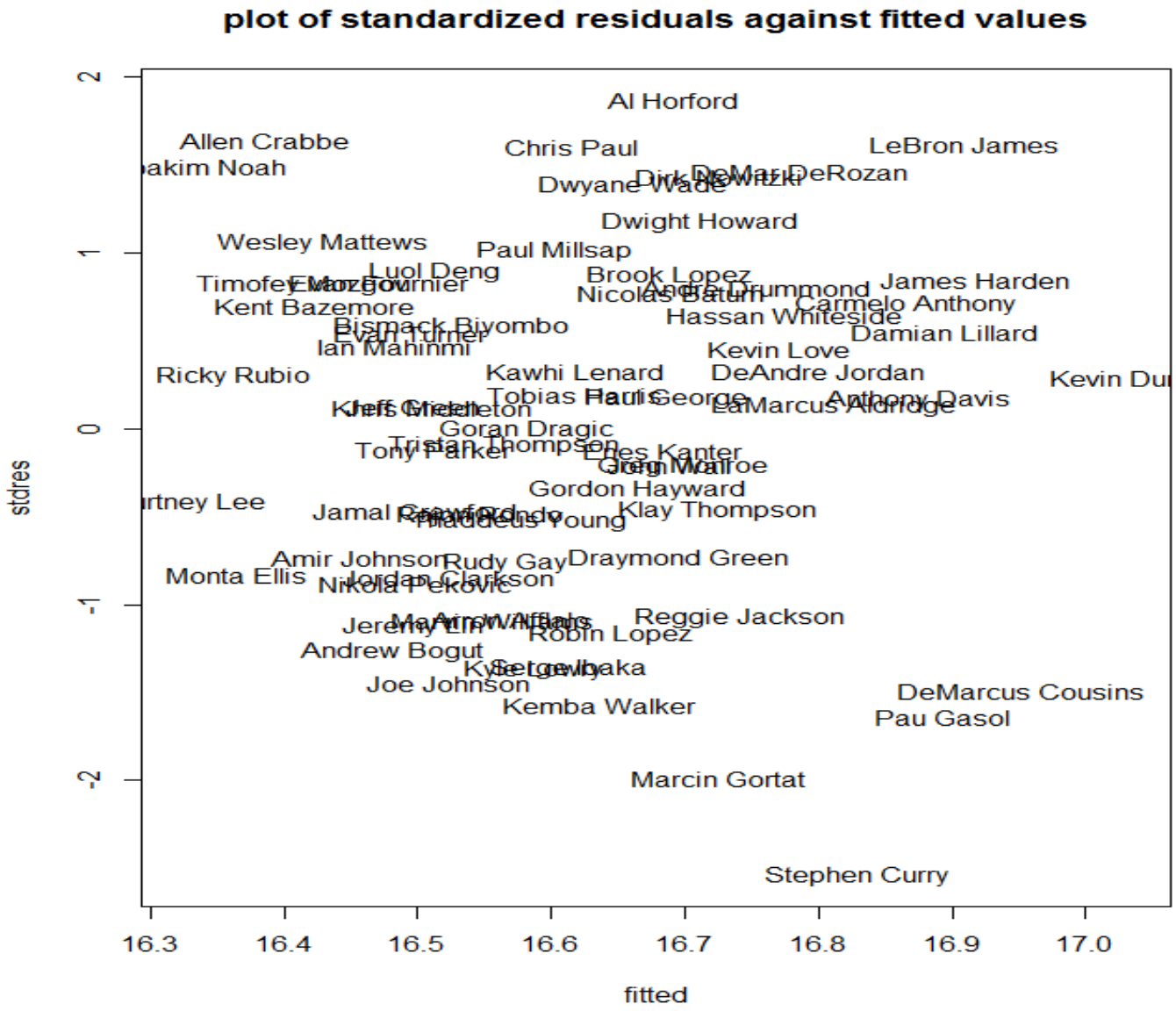
Make residual plots of your own

- You can use the functions rstandard() and rstudent() in R to get the standardized and studentized deleted residuals.

```
> stdres<-rstandard(m)
> fitted<-m$fitted
> plot(fitted,stdres,main="plot of standardized residuals against fitted values",type="n")
> text(fitted,stdres,labels=player)
> plot(fitted,stdres,main="plot of standardized residuals against fitted values",type="n")
> text(fitted,stdres,labels=Player)
```



	Player	Salary	APG	PPG	RPG	SPG
5	Al Horford	26540100	3.2	15.2	7.3	0.8
58	Stephen Curry	12112359	6.7	30.1	5.4	2.1


plot of standardized residuals against fitted values
stdres
fitted
2
1
0
-1
-2
16.3
16.4
16.5
16.6
16.7
16.8
16.9
17.0
Al Horford
Allen Crabbe
Chris Paul
LeBron James
Dwyane Wade
DeMar DeRozan
Dwight Howard
Wesley Mattews
Paul Millsap
Luol Deng
Brook Lopez
James Harden
Kent Bazemore
Carmelo Anthony
Hassan Whiteside
Damian Lillard
Kevin Love
Ricky Rubio
Kawhi Lenard
DeAndre Jordan
Kevin Du
Anthony Davis
LaMarcus Aldridge
Goran Dragic
Tony Parker
Gordon Hayward
Klay Thompson
Amir Johnson
Rudy Gay
Draymond Green
Monta Ellis
Jordan Clarkson
Nikola Pekovic
Reggie Jackson
Robin Lopez
Andrew Bogut
Joe Johnson
Kemba Walker
DeMarcus Cousins
Pau Gasol
Marcin Gortat
Stephen Curry

Detecting influential cases

- **DFITS** gives differences in fitted values when the ith case is deleted.

$$DFITS_i = \frac{\hat{Y}_i - \hat{Y}_{i(i)}}{\sqrt{MSE_{(i)} H_i}}$$

This is derived from the fact that $var(\hat{Y}_i) = \sigma^2 H_i$.

- It measures the relative change in the fitted value for the ith case when the ith case is deleted.

It can be shown that

$$\boldsymbol{DFITS_i = t_i\left(\frac{H_i}{1 - H_i}\right)^{1/2}}$$

So when will DFITS be large? Here t_i is the studentized deleted residual for the ith case.

If the absolute value of **DFITS** exceeds $2\sqrt{p/n}$ for a particular case, that case is considered influential.

You can use the function **dffits()** in R to get the DFITS values.

- The **Cook's distance** measures the difference in the predicted value when each case is deleted, but considers the impact of deleting each case on all other cases.

$$D_i = \frac{\sum_{j=1}^{n}(\hat{Y}_j - \hat{Y}_{j(i)})^2}{p \cdot MSE} = \frac{e_i^2}{p \cdot MSE}\left(\frac{H_i}{(1-H_i)^2}\right).$$

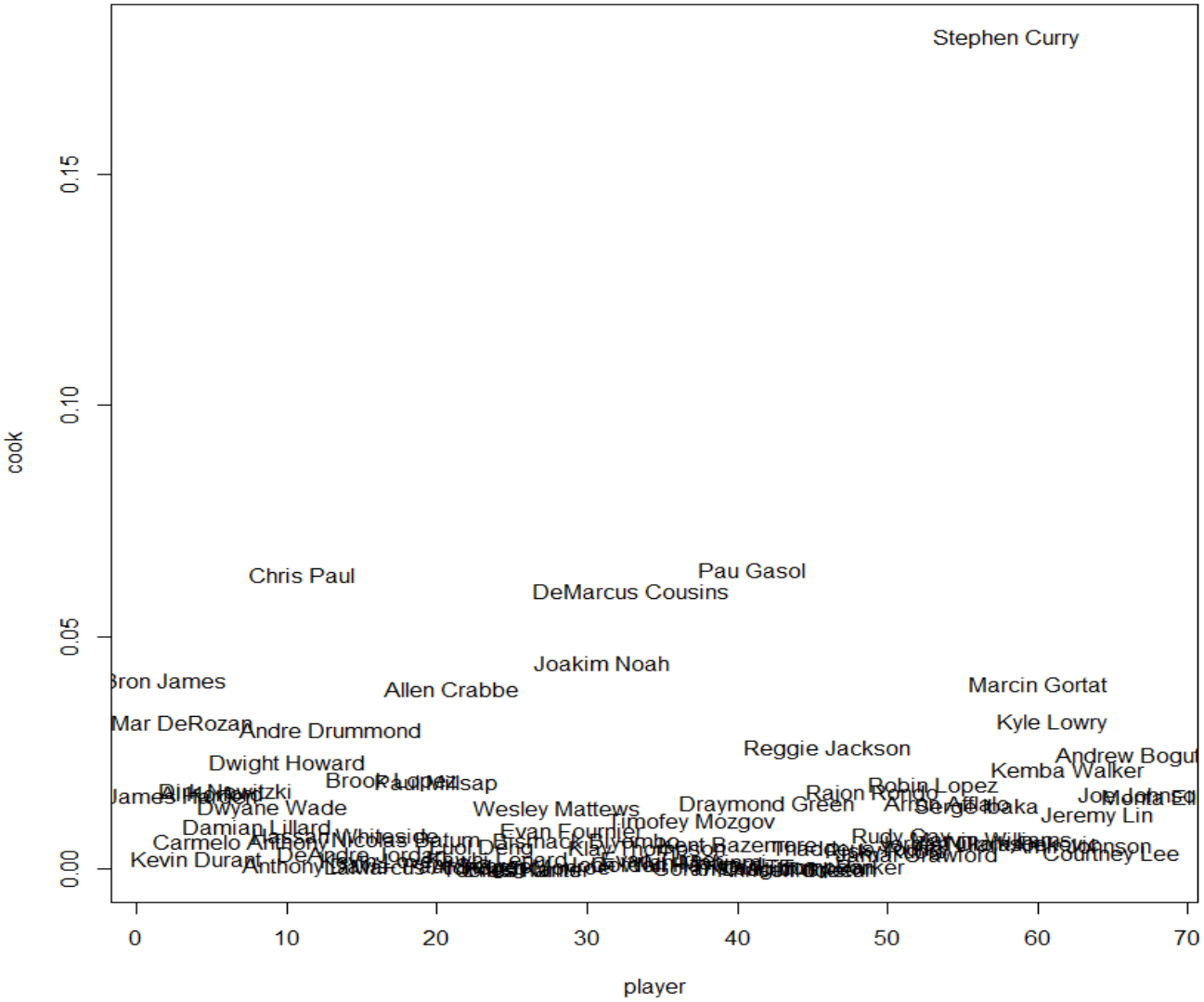
- A large Cook's distance for a particular point suggests that that point is highly influential. When will D_i be large? Here e_i is the regular residual for the ith case.

How large is large enough?

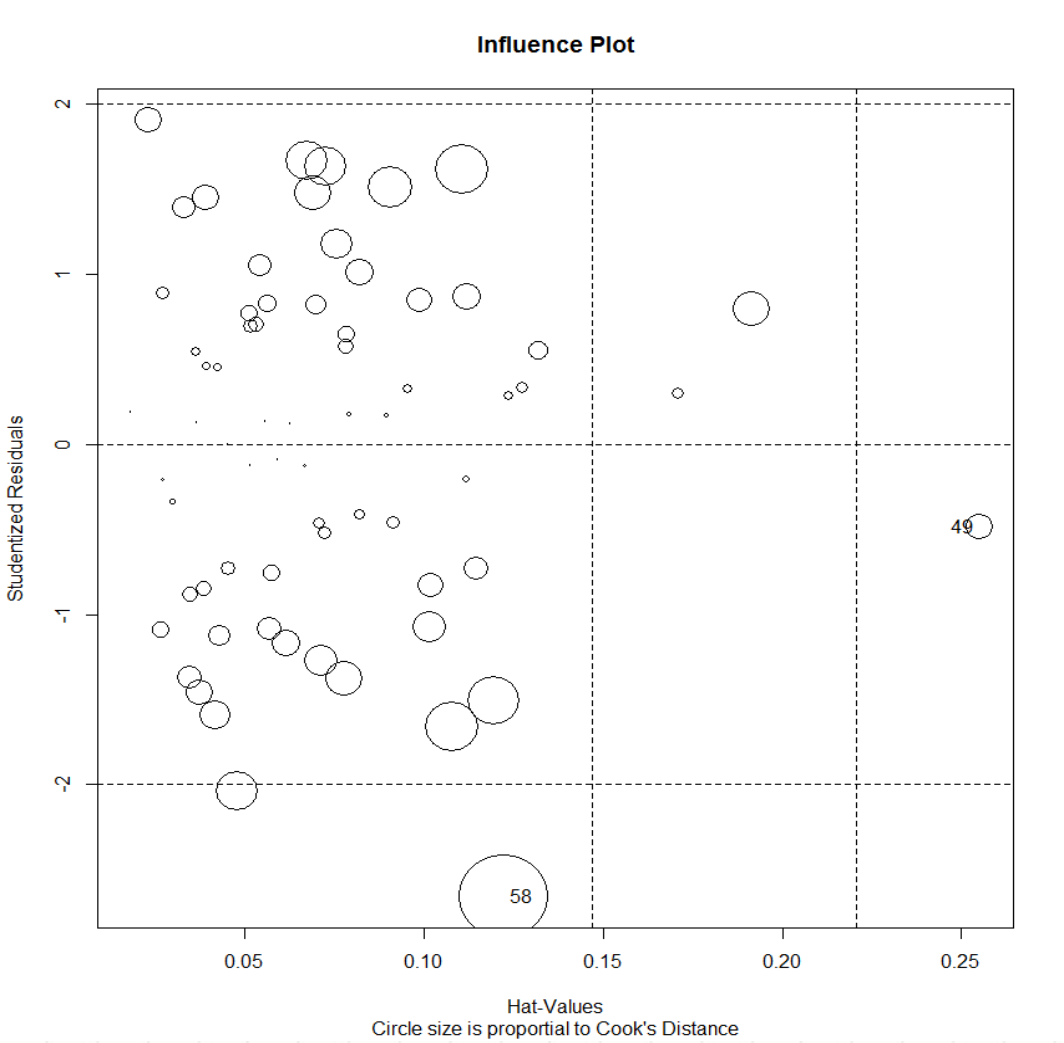
- Compare it to the F(p,n-p) distribution: if the corresponding percentile is less than about 10 or 20 percent, then the point has little influence; if the percentile is near 50 percent or more, then there is high influence.

- The best thing to do is to determine if the Cook's distance is large relative to the other n-1 Cook's distances for that sample. A value with a Cook's distance that is large relative to the other n-1 Cook's distances is considered highly influential.

```
> cook<-cooks.distance(m)
> plot(player,cook,main="plot of cook's distances",type="n")
> text(player,cook,labels=Player)
```



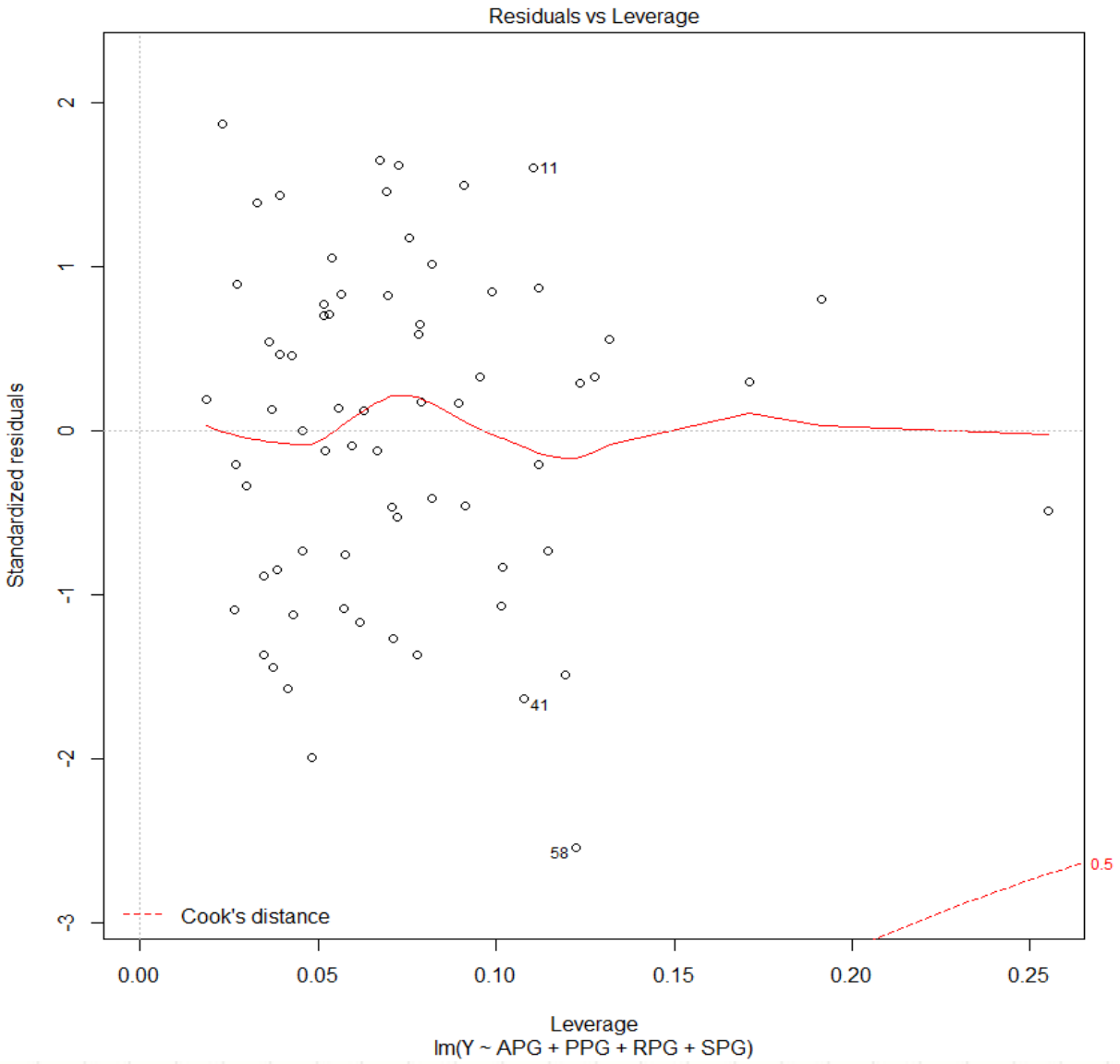
```
> library(car)
> influencePlot(m,main="Influence Plot")
```



	Player	Salary	APG	PPG	RPG	SPG
49	Rajon Rondo	14000000	11.7	11.9	6.0	2.0
58	Stephen Curry	12112359	6.7	30.1	5.4	2.1

	StudRes	Hat	CookD
49	-0.4857298	0.2550636	0.01635491
58	-2.6567842	0.1222573	0.17937978

- The function **influencePlot()** creates a “bubble” plot of Studentized residuals by hat values, with the areas of the circles representing the observations proportional to Cook's distances.
- Vertical reference lines are drawn at twice and three times the average hat value.
- Horizontal reference lines at -2, 0, and 2 on the Studentized-residual scale.




```
> Player[c(11,41,58)]
[1] Chris Paul    Pau Gasol     Stephen Curry
```